

RESEARCH ARTICLE



# Hydropower regulation reshapes littoral invertebrate communities: functional traits reveal impacts beyond richness and abundance

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## Abstract

Freshwater lake ecosystems are increasingly affected by altered water level fluctuations (WLF) from hydropower regulation, yet impacts on lake biodiversity remain understudied. We investigated how regulation influences Ephemeroptera, Plecoptera and Trichoptera (EPT) communities in boreal lakes. Three regulated and three nearby control lakes were sampled eight times across 2023–2024. We analyzed differences in abundance and species richness while accounting for temporal and spatial variability. To gain deeper mechanistic insight, we assessed community responses through analysis of composition, beta diversity-partitioning, indicator species, and functional traits. Abundance and species richness showed directional decrease in regulated lakes, although there was substantial variability among lakes. Control lakes supported compositionally similar communities with diverse functional traits, whereas communities in regulated lakes were more variable and functionally constrained. In particular, species that were temporarily attached or that required vegetation for case building were less common in regulated lakes. The observed patterns are consistent with the hypothesis that water-level fluctuations (WLF) impose stress on less mobile taxa and contribute to environmental filtering associated with reduced habitat complexity. Although responses varied among regulated lakes, our findings suggest that moderate regulation amplitudes (4–6 m) can alter littoral EPT communities. We detected clear impacts on functional composition even where changes in abundance and richness were less pronounced, highlighting its potential value as a sensitive indicator of hydropower effects in lake ecosystems.

**Keywords** Community ecology · EPT · Macroinvertebrate · Hydropower regulation · Water level fluctuations · Lake reservoir

## Introduction

Lakes are biodiverse freshwater ecosystems whose environmental conditions are increasingly altered by human activities, contributing to biodiversity loss (Reid et al. 2019). A key driver of lake ecosystem functioning is water level fluctuation (WLF), which shapes habitat availability

and ecological dynamics (Evtimova and Donohue 2016). Alterations to, for example, the amplitude, frequency, or duration of WLF can therefore negatively affect biota (Leira and Cantonati 2008). One major source of such alteration is the conversion of natural lakes into hydropower reservoirs. As global energy systems shift toward renewable sources, pressure from hydropower operations is expected to intensify due to greater demand for reservoirs as a short-term balancing energy storage to stabilize the energy grid in situations with fluctuating production from wind and solar (Vagnoni et al. 2024). Norway is expected to experience this pressure on a particularly large scale, as it holds the largest reservoir capacity in Europe, and most of its reservoirs are dammed natural lakes (Kenawi et al. 2023). Despite the growing recognition of hydropower impacts on lake biota, most existing studies rely on abundance and diversity as primary indicators. However, these metrics may be insufficiently

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sensitive to detect moderate effects of water level regulations. Community-level approaches integrating functional trait and beta diversity partitioning may offer greater resolution. This offers a mechanistically foundational base but are yet rarely applied.

Ecological consequences of expanding hydropower remain poorly understood, particularly regarding moderate amplitude WLF caused by using reservoirs as balancing storage. Recent assessments of European lakes highlight major knowledge gaps concerning the ecological effects of hydromorphological pressures (Poikane et al. 2020; Argillier et al. 2023). Comparative reviews further show that research on hydropower impacts is heavily skewed toward rivers, leaving lake ecosystems understudied and limiting effective biodiversity management in reservoirs (Williams-Subiza and Epele 2021; Krogstie et al., 2026).

Hydropower-induced WLF particularly affects the littoral zone, which is integral to lake productivity and supports diverse macroinvertebrate communities (Strayer and Findlay 2010). Among these, the insect larvae of orders Ephemeroptera, Plecoptera, and Trichoptera (EPT) serve as well-established bioindicators of disturbance in freshwater systems (Eriksen et al. 2021). EPT are considered a sensitive and ecologically informative subset of the benthic community, with a well-resolved taxonomy that allows for species-level identification. Previous studies have looked at EPT larva in relation to, e.g., hydrological regulation (Petrin et al. 2013; Evtoma and Donohue 2014), vegetation loss (Valente-Neto et al. 2021), and water quality assessments (Tubić et al. 2024). Changes in EPT communities can therefore provide insight into how and to what degree ecosystems are disturbed by regulation (Evtimova and Donohue 2014; Conroy et al. 2018). EPT also provide an important link between aquatic and terrestrial habitats, as emerging adults are an important food source for terrestrial animals such as birds (Manning and Sullivan 2021). Further, their winged adult stage allows them to travel longer distances (Arce et al. 2021), making dispersal less restricted by typical barriers for fully aquatic organisms.

Several mechanisms underlying invertebrate loss in regulated lakes have been identified. Repeated desiccation and rewetting impose physiological and behavioral stress, disproportionately affecting slow-moving or attached taxa (e.g., White et al. 2011). Degradation of littoral habitat, including erosion of fine sediment and nutrients and loss of macrophytes and detritus, further reduces resources for dependent macroinvertebrates (Carmignani and Roy 2017; Lemes da Silva and Petrucio 2018). Winter drawdowns expose littoral habitats to freezing and ice scour, reducing benthic richness and selecting for specific traits (Carmignani and Roy 2017). Yet empirical studies of this remain few, limiting our understanding of how hydropower impacts vary across biogeographical gradients and under different regulation regimes.

Changes in abundance and richness are well established simple and intuitive metrics for assessing how human activities impact fundamental community properties. However, deeper ecological insight into responses to environmental drivers is often lost in simplicity. Community-level properties can reveal mechanisms that richness-based metrics miss (e.g., Pavoine and Bonsall 2011; Fletcher et al. 2025). A growing body of literature have utilized methods for partitioning of beta diversity into turnover and nestedness originally developed by Baselga (2010). This method has promise to reveal whether environmental impacts, such as hydropower regulation, might lead to loss of species or to a turnover of species. An indicator species analysis is another tool that can be used to investigate whether any species are significantly associated with a habitat type. Lastly, functional diversity metrics has proven valuable for linking biodiversity patterns to ecosystem functioning (Gagic et al. 2015).

In this study, we investigated how hydropower regulation affects littoral EPT community structure in boreal lakes using a space-for-time substitution design. Focusing on lakes experiencing low-to-moderate regulation pressures (based on concession set maximum regulation amplitude), we sampled three regulated and three nearby control lakes in Norway during May–July of two consecutive years.

We expected that hydropower regulation would create more physically disturbed and simplified littoral habitats. Accordingly, we hypothesized that (1) EPT communities in regulated lakes would exhibit lower species richness and abundance and that (2) community composition would differ between regulated and control lakes. To explore mechanisms behind impact, we assessed whether beta-diversity was driven primarily by turnover or nestedness, and whether particular species or functional traits were disproportionately associated with regulated or control lakes.

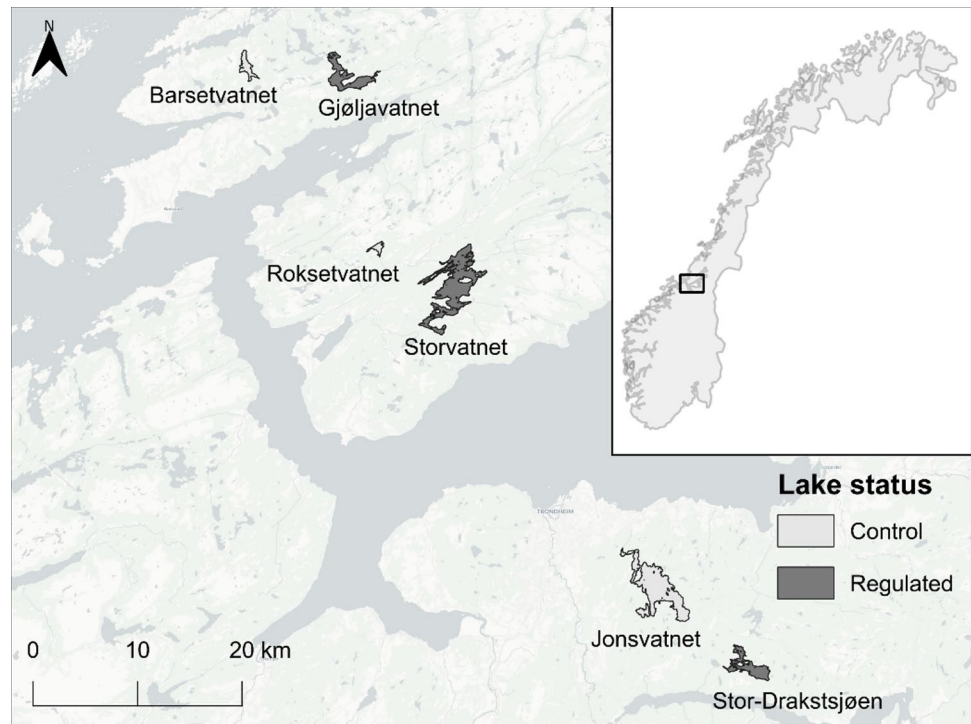
## Materials and methods

### Study region and site selection

We conducted our study in Central Norway (approx. 63°47'N to 63°17'N and 9°49'E to 10°48'E) in 2023 and 2024 from early May to early July. The study sites are located within the Atlantic biogeographical region of Trøndelag County, which has experienced an increase in precipitation over the last 20 years (Norwegian average precipitation 1400 mm/year, MET Norway 2025). In 2024, Trøndelag produced 6425 GWh of hydroelectric power, constituting 5% of national production (Statistics Norway 2025).

We selected three hydropower-regulated lakes and three nearby control lakes for a space-for time substitution design (Fig. 1, Table 1). Because only six lakes could be included

**Fig. 1** Overview of the six study lakes located in Trøndelag county, Central Norway. Data source: Norwegian Water Resources and Energy Directorate (NVE 2023b)



**Table 1** Characteristics of the study sites, data obtained from NVE (NVE 2023a, b). Average amplitude is based on measurements of max and min water levels from years with consistent monitoring (SI, Fig. S4)

| Lake status | Lake name        | Lake area (km <sup>2</sup> ) | Elevation (m.a.s.l) | Median grain size (cm) | Concession amp. (m) | Avg. amp. (m) | Concession start (year) |
|-------------|------------------|------------------------------|---------------------|------------------------|---------------------|---------------|-------------------------|
| Control     | Barsevatnet      | 1.7                          | 60.0                | 2.47                   | –                   | –             | –                       |
|             | Roksetvatnet     | 0.7                          | 193.0               | 6.37                   | –                   | –             | –                       |
|             | Jonsvatnet       | 14.3                         | 149.4               | 1.22                   | –                   | –             | –                       |
| Regulated   | Gjølvatnet       | 5.2                          | 51.4                | 2.72                   | 4.0                 | 3.3           | 1950                    |
|             | Storvatnet       | 15.5                         | 131.9               | 2.20                   | 5.9                 | –             | 1956                    |
|             | Stor-Drakstsjøen | 4.5                          | 265.6               | 1.90                   | 5.0                 | 3.2           | 1919                    |

in the study, considerable effort was made to minimize confounding factors when selecting study sites. The lakes were selected based on their overall similarity in physical and chemical conditions, their exposure to the same overall climatic conditions due to spatial proximity, absence of known confounding disturbances (i.e., rotenone treatments, agricultural runoff), and logistical considerations (Supplementary Information (SI), Fig. S1, Table S1). Still, finding suitable control lakes of similar size was challenging, as many nearby lakes were either regulated or faced other disturbances. The regulated lakes were selected to have a “moderate” concession set maximum regulation amplitude of 4–6 m. About 15% of Norwegian reservoirs experience similar regulation, which is below the national median of 7 m (SI, Fig. S2). When selecting a study site at each lake, we chose locations in the upper stony littoral, which predominantly resembled

exposed lakeshores with small gravel/stony substrate (median grain size 1.22–6.37 cm). Fish known to predate on benthic invertebrates are present in all lakes, mainly Brown trout (*Salmo trutta*) and Arctic charr (*Salvelinus alpinus*) (SI, Table S2).

### Macroinvertebrate sampling

The sampling design of this project was created to (1) catch the spring–summer benthic macroinvertebrate community composition of each lake and (2) catch the emergence timing for selected insects that is analyzed elsewhere. Sampling of macroinvertebrates was therefore performed biweekly from ice melt in 2023 (5 May) and 2024 (7 May), four events per year, using kick-sampling and malaise-traps. In this

study, we use only benthic macroinvertebrate data from kick samples.

Sampling of benthic macroinvertebrates was conducted at one upper stony littoral zone site per lake at approximately 0.4 m depth using a standardized kick-sampling method (Frost et al. 1971). During each 1-min sampling event, approximately 1 m<sup>2</sup> of the top substrate was disturbed by kicking and loosened benthic invertebrates and organic material was collected by continuously sweeping with a kick net (25 × 25 cm frame, 0.25 mm mesh). Multiple replicates (five in 2023, three in 2024) were taken per sampling event, each taken parallel to the shoreline spaced 1–2 m apart to avoid disturbance between replicates. Sampling depth was kept constant, but in situations with water level fluctuations, the sampling station was moved perpendicular to the shoreline within the upper stony littoral.

### Identification and EPT traits

In the lab, all EPT were sorted from the sample and identified to the lowest possible taxonomic level with the support of the following literature: Ephemeroptera (Nilsson 1996), Plecoptera (Lillehammer 1988), and Trichoptera (Rinne and Wiberg-Larsen 2017). Identified material is preserved in the NTNU University Museum collections and data can be accessed through the Global Biodiversity Information Facility (GBIF; Hårsaker et al. 2025).

To manage sample processing time across the 48 sampling events, one replicate was processed per event, with the exception of the first event where three replicates were processed. This initial processing of a higher number of replicates was done to compare variation in species richness and abundance between replicates within an event with the among-event variation across all lakes. The coefficient of variation among replicates within events was consistently lower than the variation among sampling events within lakes, indicating that temporal variability exceeded within-event sampling variability. We therefore kept the single replicate protocol for the remaining samples, allowing us to process eight events per lake over 2 years. Selection of each replicate was done at random prior to retrieving the material from storage to avoid selection bias.

A number of functional traits can be described for EPT, and several may influence invertebrate survival and fitness in relation to hydropower regulation. When selecting traits for analysis, we considered the cost of obtaining the trait data, associated uncertainty regarding species-level information, and the relevance of traits based on previous hydropower-impact studies. Based on the mentioned factors, we investigated the two traits locomotion and substrate relation (burrower, crawler, temporarily attached, full water swimmer) and armor (sclerotized, silk, mineral, mixed, vegetation). These traits are closely related to species morphology and

can thus be accurately described for all species in our dataset. Both traits are expected to influence the vulnerability to physical disturbance from water level fluctuations. Locomotion and substrate relation determines a species ability to track fluctuating water lines. Armor reflects the species dependence on specific substrates. For locomotion and substrate relation, we based categorization on Tachet et al. (2010) and assigned each species to its main trait category. For most species, trait data could be obtained from the database [freshwaterecology.info](https://freshwaterecology.info) (Schmidt-Kloiber and Hering 2015). For armor, the categories reflect the outermost protective armoring of each species, which could be cases built from different materials or a sclerotized exoskeleton. For Trichoptera, we followed descriptions in Rinne and Wiberg-Larsen (2017), while all Ephemeroptera and Plecoptera were considered sclerotized.

### Environmental data

To control our assumptions regarding similar climatic conditions across lakes and differing water-level-fluctuation regimes between control and regulated lakes, we measured water temperatures and water levels. Both assumptions were met; see SI (Fig. S3–5) for details. Water temperature was recorded every 6 h using temperature data loggers (HOBO Pendant, Onset Computer Corp., Bourne, Massachusetts, USA). Water levels were measured throughout the season in control lakes by measuring depth against a marked wooden post, which was converted to m.a.s.l. using a satellite antenna (Trimble GNSS, Trimble Inc., Westminster, Colorado, USA). Openly accessible information on hydropower concessions and geographic lake data were sourced from the Norwegian Water Resources and Energy Directorate (NVE), through the map service NVE Atlas (<https://atlas.nve.no>) or the NVE thematic data site (<https://nedlasting.nve.no/gis/>). Hydrological data for the regulated lakes Gjøljavantet and Stor-Drakstjøen were openly accessible from the NVE web portal Sildre (<https://sildre.nve.no/>).

### Statistical analyses

All analyses were conducted in the statistical computing language R (v4.2.2, R Core Team 2022). As several analyses required species-level resolution, we kept only EPT occurrences identified to species-level to ensure consistency in the underlying dataset used throughout the manuscript. Out of the total number of EPT individuals found across the 60 analyzed samples ( $n = 7956$ ), 33% could be identified to species level. While all data points could have been used for analyses of abundance, species-level abundances may provide a more stable representation of the communities—less influenced by possible differences in recruitment pulses (14% of individuals were taken only to order). Based on experience

from, e.g., the Swedish lake monitoring programme (Fölster et al. 2014), sampling later in the season would likely have resulted in a higher percentage of individuals identified to species level. Our analysis choices thereby reflect a prioritization of characterizing the early season species-level community structure.

To graphically assess whether the communities were adequately sampled, we constructed rarefaction curves using the *vegan* package (v2.6–4, Oksanen et al. 2022), adding samples in random order. The curves had begun leveling out at ten samples, suggesting that the majority of species in the communities were identified (Fig. 2). Further, to evaluate whether natural spatial turnover contributed to differences in community composition, we tested for a relationship between geographic distance and Bray–Curtis community dissimilarity using a Mantel test with 9999 permutations.

Prior to multivariate analyses, abundance or presence–absence data from all sampling events within each lake were aggregated into a single community matrix, resulting in one observation per lake. Consequently, the lake constituted the experimental unit for permutational multivariate analysis of variance (PERMANOVA),  $\beta$ -diversity partitioning, and indicator species analyses. Because the analyses were conducted on independent lake-level observations, unrestricted permutations were used.

### Abundance and species richness

To investigate whether hydropower regulation affected EPT abundance and species richness, we fitted generalized linear mixed-effects models (GLMMs) using the package *lme4* (v1.1–31, Bates et al. 2015) and performed model selection

based on Akaike information criterion corrected for small sample sizes (AICc) (Burnham and Anderson 2004). For AICc-based model selection, all models with an AICc-score of  $< 2$  were considered as having substantial support. Model support could be further evaluated by comparing evidence ratios (i.e., how many times more likely is model 1 than model  $n$ , calculated as  $w_{\text{model 1}}/w_n$ ). The predictor variables of interest were regulation status (*status*; regulated, control) and year (*year*; 2023, 2024) which was included to control for the differences in spring temperatures between years (SI, Fig. S5).

Abundance was investigated using mixed-effects models with negative binomial distributions, as data showed strong overdispersion (mean = 43, var = 2813). First, the dispersion parameter (theta = 1.33) was extracted from the full model. Then, all relevant candidate models were fitted with a fixed theta, and model selection was performed with the *MuMIn* package (v.1.47.1, Bartoń 2022). Supported models ( $\Delta\text{AICc} < 2$ ) were finally refitted without a fixed theta.

Full model : abundance  $\sim$  status \* year + (1|locality)

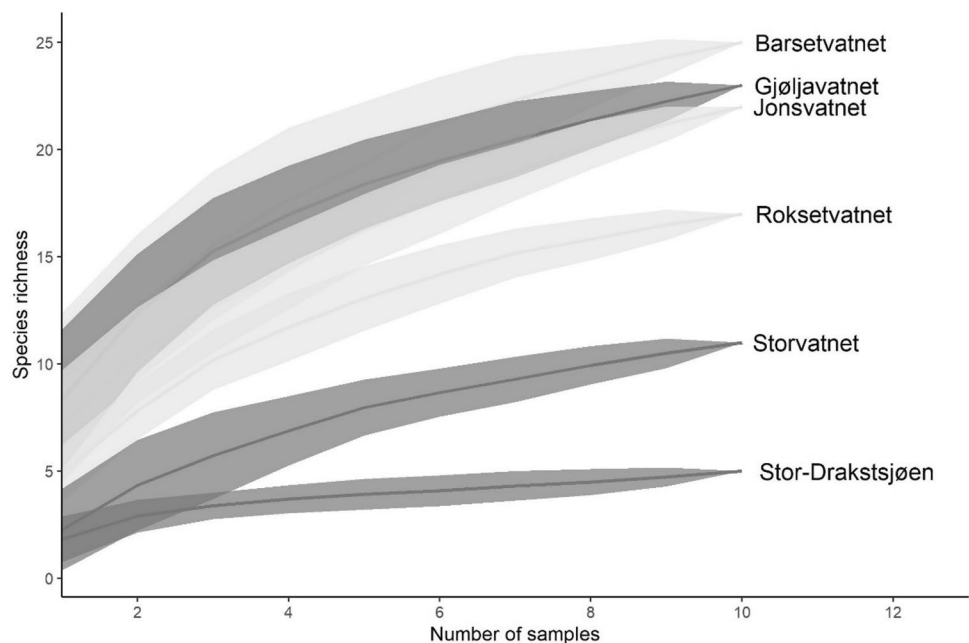
Species richness was investigated using mixed-effects models with a Poisson distribution, as data represented species counts with only moderate overdispersion (mean = 6, var = 13).

Full model : species richness  $\sim$  status \* year + (1|locality)

### Differences in community composition

We tested for differences in composition of EPT communities between regulated and control lakes by first creating site

**Fig. 2** Sample-based rarefaction curves for each study lake showing mean and standard deviations calculated from random permutations of the data. Control lakes in light gray and regulated lakes in dark gray



x species abundance matrices for each lake and pooling all samples per lake, followed by the calculation of pairwise distance matrices between all lakes using Bray–Curtis dissimilarity indices. We then performed a PERMANOVA (999 permutations) using the function `adonis2` from the `vegan` package. Homogeneity of group dispersions (PERMDISP) was tested using functions `betadisper` on the distance matrix followed by `permutest` from the same package. To visualize the PERMANOVA results, we performed a two-dimensional non-metric multidimensional scaling (NMDS) using function `metaMDS` from the `vegan` package.

### Partitioning of $\beta$ -diversity

To assess whether potential changes in community composition due to hydropower regulation is driven by species loss or replacement, we performed a  $\beta$ -diversity partitioning analysis using the `betapart` package (v.1.6.1, Baselga and Orme 2012). Using a site x species presence-absence matrix with pooled samples, we calculated pairwise  $\beta$ -diversity based on Jaccard dissimilarity ( $\beta_{jac}$ ) and its partitioning into nestedness ( $\beta_{jne}$ ) and turnover ( $\beta_{jtu}$ ). We repeated the PERMANOVA (999 permutations) on each dissimilarity component and tested for significant differences in dispersion (PERMDISP) for each component as described above.

### Indicator species analysis

An indicator species analysis was performed to investigate whether any species are significantly associated ( $p \leq 0.05$ ) with either regulated lakes or control lakes. The analysis was conducted using the `indicspecies` package (v1.8.0, De Cáceres and Pierre 2009), with samples pooled to a site x species abundance matrix and 999 permutations.

### Difference in functional traits

To test for differences between regulated and control lakes in the abundance of EPT individuals belonging to trait levels related to locomotion and surface relation and armor, we fitted generalized mixed-effects models with negative binomial distributions due to data overdispersion using the R package

`glmmTMB` (v1.1.9, Brooks et al. 2017). Each trait model included an interaction between the predictor variables regulation status (*status*; regulated, control) and trait (*locomotion and surface relation*; burrower, crawler, temporarily attached, full water swimmer; and *armor*; sclerotized, silk, mineral, mixed, vegetation). A random effect structure was included to account for repeated measures within localities and events. Following model fitting, we used estimated marginal means to evaluate differences in abundance between control and regulated lakes within each trait level. Pairwise contrasts were calculated, and  $p$ -values were adjusted for multiple testing using the Holm method.

Full model :  $\text{abundance} \sim \text{status} * \text{trait} + (1|\text{locality}/\text{event})$

## Results

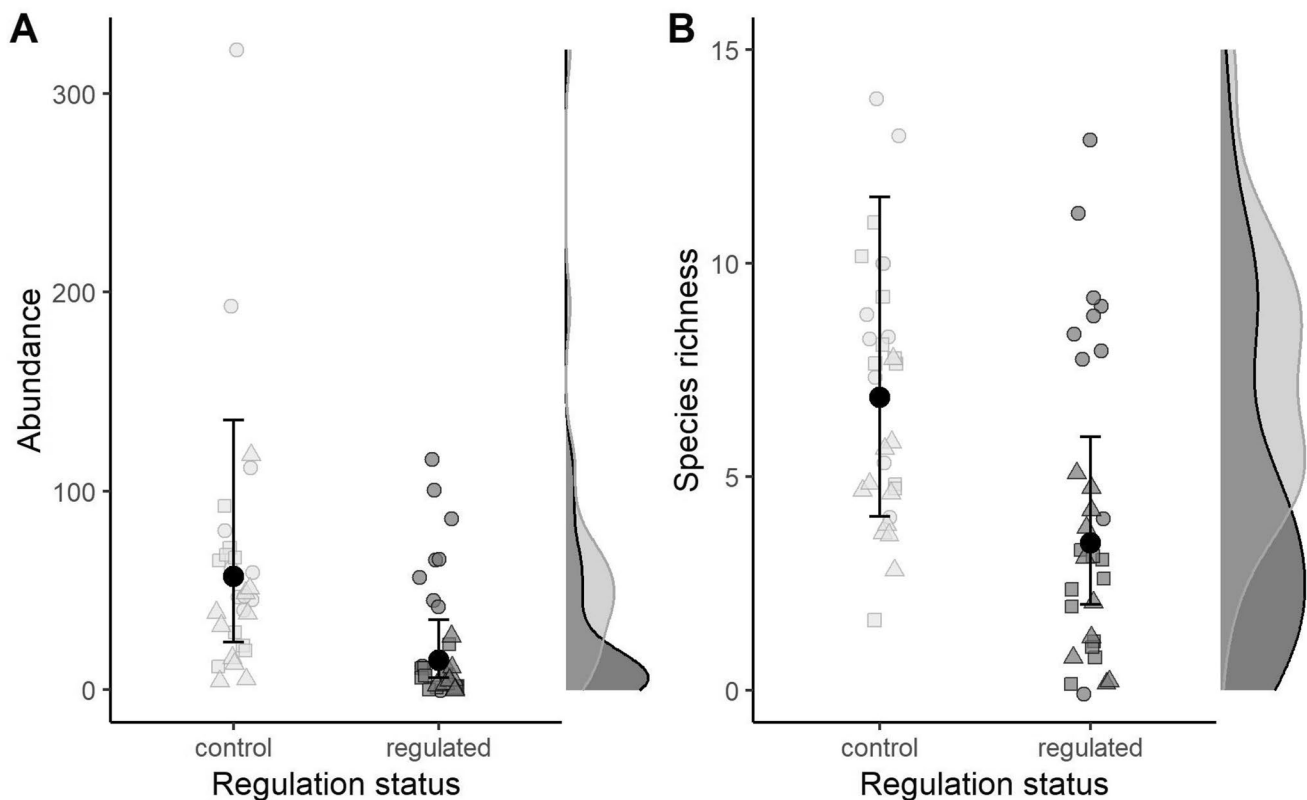
A total of 2607 EPT individuals could be identified to species level, with the number per sample ranging from 0 to 322 with an average of 43.5 (median = 30.5; SD 53.0); lake-wise summary statistics are found in Table 2. We identified 45 different species across all study lakes, of which 15 belonged to the order Ephemeroptera, 3 to Plecoptera, and 27 to Trichoptera. The full species list can be found in Supplementary Information (Table S3). Geographic distance was not significantly related to community dissimilarity (Mantel  $r = 0.13$ ,  $p = 0.25$ ), suggesting that community dissimilarity did not systematically increase with geographic distance across the study area.

### Abundance

Model selection based on AICc indicated that the model including lake status received the greatest support (SI, Table S4,  $w_i = 0.43$ ; Fig. 3A). Two other models also received substantial support ( $\Delta\text{AICc} < 2$ ): the intercept-only model ( $\Delta\text{AICc} = 1.19$ ) and the model including status and year ( $\Delta\text{AICc} = 1.78$ ). Evidence ratios suggested that the best model was about two times more likely than the second- and third-ranked models. In the top-ranked model, abundance

**Table 2** Summary statistics for EPT abundance and species richness, based on 2670 individuals from 10 samples collected during 8 events

| Lake status | Lake name        | Abundance |      |     |     | Species richness |     |     |     |
|-------------|------------------|-----------|------|-----|-----|------------------|-----|-----|-----|
|             |                  | Mean      | SD   | Min | Max | Mean             | SD  | Min | Max |
| Control     | Barsetvatnet     | 100.6     | 90.6 | 40  | 322 | 8.8              | 3.2 | 4   | 14  |
|             | Roksetvatnet     | 36.5      | 33.4 | 4   | 118 | 5.0              | 1.4 | 3   | 8   |
|             | Jonsvatnet       | 50.2      | 27.1 | 12  | 92  | 7.4              | 2.7 | 2   | 11  |
| Regulated   | Gjøljavatnet     | 59.0      | 36.5 | 0   | 116 | 7.9              | 3.6 | 0   | 13  |
|             | Storvatnet       | 7.5       | 8.2  | 0   | 27  | 2.5              | 2.0 | 0   | 5   |
|             | Stor-Drakstsjøen | 6.9       | 7.0  | 0   | 23  | 1.9              | 1.1 | 0   | 3   |



**Fig. 3** Effect plots for highest ranked mixed models predicting (A) EPT abundance and (B) EPT species richness, as a function of regulation status. Predicted mean value is presented along with its 95%

confidence interval. Raw data shown as jitter points (circle, Barsevatnet/Gjøljavatnet; triangle, Roksetvatnet/Storvatnet; square, Jonsvatnet/Stor-Drakstsjøen) and summarized by marginal density plots

was lower in regulated lakes compared with control lakes (SI, Table S5,  $\log_e \beta = -1.35$ ), corresponding to an approximately 33% reduction in expected abundance. However, the 95% confidence interval slightly overlapped zero ( $\log_e$  95% CI  $-2.82$  to  $0.09$ ), indicating uncertainty around the magnitude of the effect. Investigating model performance revealed that much of the variance was explained by differences between lakes (marginal  $R^2 = 0.300$ , conditional  $R^2 = 0.627$ ).

### Species richness

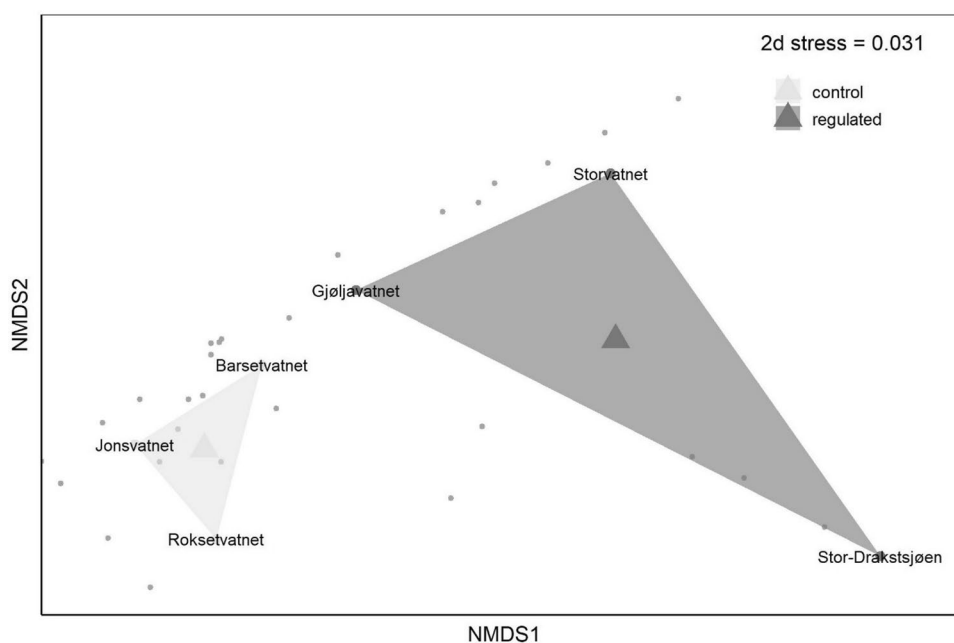
For species richness, the AICc-based model selection also gave substantial support to three models, with the model including lake status receiving the greatest support (SI, Table S4,  $w_i = 0.39$ ; Fig. 3B). While evidence ratios suggested that the second ranked intercept-only model was almost as likely as the highest ranked model ( $\Delta\text{AICc} = 0.43$ , ratio = 1.2), the third ranked model including status and year received considerably less support ( $\Delta\text{AICc} = 1.99$ , ratio = 2.7). In the top-ranked model, species richness was lower in regulated lakes compared with control lakes (SI, Table S5,  $\log_e \beta = -0.69$ ), corresponding to an

approximately 36% reduction in expected richness. However, the 95% confidence interval overlapped zero ( $\log_e$  95% CI  $-1.60$  to  $0.19$ ), indicating substantial uncertainty in the magnitude and direction of the effect. In addition, here much of the variance was explained by lake differences (marginal  $R^2 = 0.251$ , conditional  $R^2 = 0.663$ ).

### Differences in community composition

The PERMANOVA did not support any significant differences in community composition between regulated and control lakes ( $F_{1,4} = 1.91$ ,  $R^2 = 0.32$ ,  $p = 0.10$ ). However, dispersion differed significantly between regulated and control lakes (PERMDISP:  $F_{1,4} = 14.34$ ,  $p = 0.0014$ ), indicating greater within-group variability in community composition among the regulated lakes. We visualized the communities using a two-dimensional NMDS plot (Fig. 4). The two-dimensional (2D) stress was less than 0.05 (2D stress = 0.031), which following Clarke (1993), can be interpreted as a very good fit with no prospect of misinterpretation.

**Fig. 4** Two-dimensional NMDS of the EPT invertebrate community dissimilarity in the study lakes, gray points mark individual species



### $\beta$ -diversity partitioning

Beta diversity partitioning indicated that species composition differed between regulated and control lakes (mean Jaccard dissimilarity = 0.755, Table 3). The turnover component accounted for approximately 70% of this dissimilarity, whereas the nestedness component contributed around 30%. However, PERMANOVA did not detect a significant difference between groups (PERMANOVA;  $p > 0.05$  for  $\beta_{jac}$ ,  $\beta_{jne}$ , and  $\beta_{jtu}$ ). Dispersion tests indicated that, when only considering species presence/absence, within-group variability was similar (PERMDISP;  $p > 0.05$  for  $\beta_{jac}$ ,  $\beta_{jne}$ , and  $\beta_{jtu}$ ). Furthermore, total beta diversity was high among both control ( $\beta_{jac} = 0.675$ ) and regulated ( $\beta_{jac} = 0.787$ ) lakes. While turnover constituted the largest component of beta diversity in both groups, within-group partitioning showed a minimal contribution of nestedness among control lakes ( $\beta_{jne} = 0.078$ ,  $\approx 12\%$ ), but a substantially greater contribution among regulated lakes ( $\beta_{jne} = 0.307$ ,  $\approx 39\%$ ). Thus, although turnover remained the dominant component, nestedness contributed more strongly to community dissimilarity among the regulated lakes.

### Indicator species analysis

No significant indicator species were identified for either regulated or control lakes.

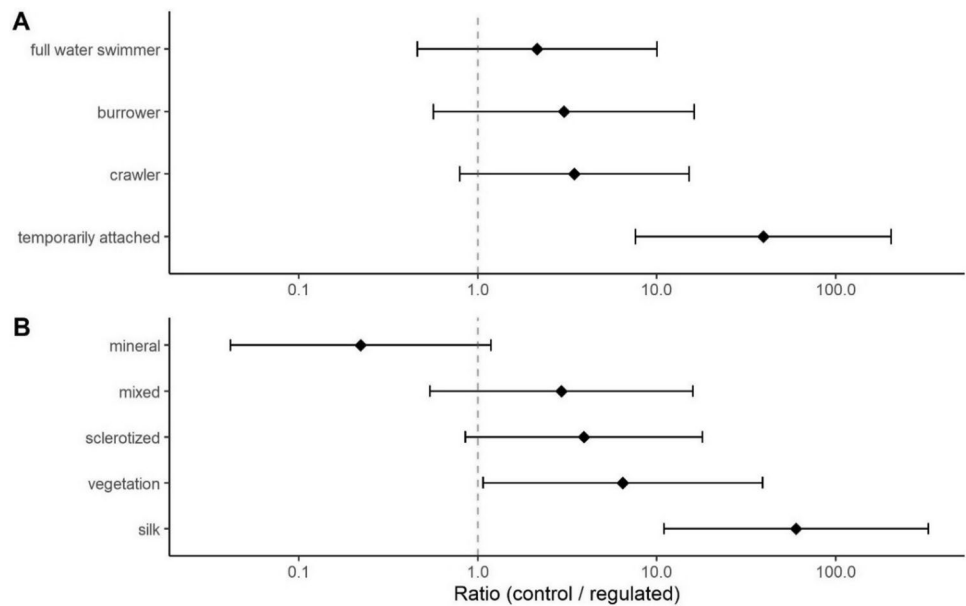
### Functional traits

The analysis of functional traits revealed that certain types of locomotion and surface relation and protective body armor were more common in control lakes than in regulated lakes (Fig. 5; SI, Table S6). While control lakes tended to have higher abundances across most trait types (ratios  $> 1$ , estimated from negative binomial GLMMs), this difference was only statistically supported for temporarily attached individuals (ratio = 39.4, 95% CI 7.61–204), as well as case builders using vegetation (ratio = 6.47, 95% CI 1.07–39.1) and silk (ratio = 60.1, 95% CI 11–329). Only mineral case builders were more common in regulated lakes (ratio = 0.22, 95% CI 0.04–1.18), although this difference was not statistically supported.

**Table 3** Taxonomic  $\beta$  diversity and its components turnover and nestedness calculated for all lakes, within and between control and regulated lakes. Calculations were done with the Jaccard dissimilarity index, and between-group calculations are mean values

| Comparison                        | <i>N</i> | $\beta$ total | Turnover | Nestedness | % $\beta$ turnover |
|-----------------------------------|----------|---------------|----------|------------|--------------------|
| Between control and regulated     | 6        | 0.755         | 0.529    | 0.226      | 68.3               |
| All lakes                         | 6        | 0.832         | 0.716    | 0.116      | 86.0               |
| Within control lakes              | 3        | 0.675         | 0.597    | 0.078      | 88.5               |
| Within hydropower-regulated lakes | 3        | 0.787         | 0.480    | 0.307      | 61.0               |

**Fig. 5** Forest plot showing the abundance ratio and log-scale 95% CIs for each functional trait level for (A) locomotion and surface relation and (B) armor. Ratios were derived from model-based estimated marginal means, with values greater than 1 indicating higher abundance in control lakes



## Discussion

In this study, we hypothesized that EPT species richness and abundance would be lower in regulated lakes compared with nearby lakes unaffected by hydropower, and that there would be differences in community composition. Our findings indicate lower abundance and species richness in regulated lakes compared with control lakes, but statistical support was not conclusive. Lake community composition clustered by regulation status, but with limited statistical support. Functional trait analyses revealed significant differences in the prevalence of several trait types between regulated and control lakes. Our findings suggest that the impact of hydropower regulation on EPT communities is more strongly reflected in community structure than in the traditional metrics of abundance and richness.

Our results show more variability than expected in abundance, species richness, and community composition within the regulated lake group. We consider two nonmutually exclusive explanations that may account for this pattern: (1) there may be unaccounted-for differences in lake-specific characteristics and (2) there may be differences in impact intensity from hydropower regulation. We here interpret our results in light of these two possible explanations and discuss possible mechanisms behind both the observed differences between EPT communities in control and regulated lakes as well as the observed variability within the regulated lake group.

### Directional decrease in abundance and species richness

Abundance and species richness showed directional decrease in regulated lakes. This was, however, associated with

considerable uncertainty, and statistical support for compositional differences between groups was not conclusive. This likely reflects within-group heterogeneity among the regulated lake communities. While Storvatnet and Stor-Drakstsjøen exhibited low abundance and species richness, the third lake (Gjøljavatnet) maintained values comparable to control lakes (see Table 2 and Fig. 3).

Previous studies have reported loss of taxa richness in lakes experiencing similar hydropower regulation pressures to the ones in this study. Moreover, the threshold for impact seem to vary among systems. While clear reductions in taxa richness were observed from 5 m regulation amplitude in a Scottish study (Smith et al. 1987), studies from boreal lakes suggest that impacts can occur at lower levels (2–3 m), potentially due to additional stressors such as freeze exposure (Aroviita and Hämäläinen 2008; White et al. 2011). Our findings from the regulated lake Gjøljavatnet therefore diverge from previous studies, as no clear reduction in richness is observed at a concession-given amplitude of 4 m. Regarding abundance, previous findings were more variable. While Aroviita and Hämäläinen (2008) found no significant relationship between abundance and regulation pressure, a Canadian study (Trottier et al. 2019) found that abundance decreases with increasing amplitude (range 0.3–7.2 m). We consequently note the uncertainty surrounding expected responses in abundance, particularly at lower regulation amplitudes.

A possible mechanism leading to a directional decrease in abundance and species richness could be that altered WLF causes stress/disturbance, which directly impacts invertebrate survival. This is consistent with the observed reduction of temporarily attached species, e.g., Trichoptera larva belonging to the family Polycentropodidae. The apparent

loss of Polycentropodidae following regulation has been reported by others (Aroviita and Hämäläinen 2008) and is commonly thought to be a result of either their reduced ability to move or the metabolic cost of moving. An alternative explanation could be that hydropower infrastructure decreases watershed connectivity and isolates lake populations, potentially increasing extinction risk. This fragmentation effect has been documented in fish populations (Barbarossa et al. 2020; Wilson et al. 2022). Nevertheless, we argue that environmental factors are likely to cause the observed differences, not dispersal constraints, as EPT species are able to cross barriers in their adult terrestrial stage. This has previously been shown for distribution and assembly of Trichoptera species in mountain lakes (de Mendoza et al. 2015).

Sampling only in spring and summer may have led to an underreporting of species richness, as instars of fall-emerging species can be too small to identify early in the season. However, we expect this underreporting of species to be consistent across lakes, not skewing the direction of results, as both sampling time and environmental conditions were similar. Moreover, spring–summer is a period with high energy demands of terrestrial predators in boreal and temperate regions, and EPT species represent a substantial biomass subsidy to terrestrial consumers, including aerial insectivore birds and bats (de Jong and Ahlén 1991; Bartrons et al. 2013; Manning and Sullivan 2021). Our findings therefore suggest that hydropower regulation may reduce aquatic subsidy available to dependent terrestrial consumers during a critical period of the year. This underlines the importance of considering cross-ecosystem linkages when evaluating the full ecological footprint of hydropower regulation.

### Community responses may reflect an impact gradient

Throughout our analyses, regulated lake communities showed increasing divergence from control communities, consistent with an impact gradient. While analyses of abundance and species richness suggested that one regulated lake (Gjøljavatnet) was seemingly unaffected by hydropower (SI, Fig. S6), the NMDS ordination positioned its community between the control lakes and the other regulated lakes. Functional trait analyses further support the idea of a gradient in response rather than a uniform effect of regulation. While all regulated lakes displayed the same shifts in trait abundances, particularly reduced abundances of temporarily attached silk case builders and vegetation-dependent case builders, only Gjøljavatnet maintained high trait diversity similar to controls (SI, Fig. S7, S8).

Both studies by Aroviita and Hämäläinen (2008) and White et al. (2011) from boreal systems found that community composition in regulated lakes differed from

reference/natural lakes. They further reported that compositional changes often appeared even at relatively small fluctuations, becoming more pronounced with increasing disturbance. Our findings are consistent with this pattern.

Possible mechanisms leading to an impact gradient could be that the regulated lake communities experience different levels of disturbance from WLF, either due to differences in operation regimes or unmeasured lake-level characteristics that modify impact. One aspect of the operation regime known to affect the level of disturbance on communities is amplitude. Differences in concession-set amplitudes between the regulated lakes (Table 1) are in line with this explanation. Regulation pressures in Gjøljavatnet may not be crossing a critical amplitude threshold, allowing for the maintenance of biodiverse benthic communities. However, investigating publicly available WLF data (SI, Fig. S4) contradicts this explanation, as the realized regulation amplitude in the last decades in both Gjøljavatnet and Stor-Drakstsjøen are similar. There could also be other operational differences that we are unable to detect due to limited access to data considered proprietary business information.

Another possibility is that lake slope modifies the level of disturbance on invertebrate communities. The sampling site in Gjøljavatnet had a steeper slope than the sites in the other regulated lakes (see SI, Fig. S1), which likely reduces the extent of habitat exposed to desiccation and freezing. Previous studies support that such differences in morphometry can affect WLF impact (Carmignani and Roy 2017). Some macrophytes and aquatic mosses were also observed in Gjøljavatnet, possibly maintained by the steeper slope, which is known to support biodiversity of macroinvertebrates in boreal lakes (Heino 2008). In their study of determinants for the vertical extension of helophyte zones in Finnish lakes, Keto et al. (2018) found that after WLF, slope was one of the key factors explaining helophyte extent. Their results indicated that there was an optimal slope range, with an initial increase in extent with slope until a threshold was reached. Other studies have similarly linked macrophyte cover and biomass to slope, with variation in optimum conditions also depending on, for example, wind exposure and light penetration (see for instance Duarte and Kalff 1986). A more extensive study of the lakeshore morphology and macrophytes present at our study sites would be needed to determine how optimal slopes are for macrophyte growth.

### Potential species turnover associated with hydropower regulation

Our results further suggest that hydropower regulation may contribute to species turnover, as indicated by the  $\beta$ -diversity partitioning and supported by differences in functional trait composition. The absence of a significant

distance–decay relationship suggests that broad-scale spatial turnover was limited across the study area. We therefore interpret the observed  $\beta$ -diversity and functional trait patterns in the context of lake-specific characteristics and differences in hydropower regulation rather than regional geographic trends.

Partitioning of beta diversity into turnover and nestedness shows that turnover was high when comparing control and regulated lakes. This suggests that differences between the groups could be the result of species replacement. While some studies comparing beta diversity partitioning between disturbed and reference habitats also found turnover to be high (Peralta et al. 2018; Petsch et al. 2021), other studies found the nestedness component to dominate (Wang and Han 2023). Similarly to Hawkins et al. (2015), we observed slightly higher total beta diversity among regulated lakes than among controls, contrary to our predictions of decreased habitat complexity in regulated lakes. Within-group partitioning revealed minimal nestedness among control lakes, but substantially higher nestedness among regulated lakes. This pattern suggests that EPT communities in regulated lakes Storvatnet and Stor-Drakstsjøen may represent subsets of the community in Gjøljavatnet, strengthening the idea of an impact gradient.

Predicting how beta diversity changes can be challenging and highly context-dependent (Godsoe et al. 2022). While anthropogenic disturbances are commonly believed to lead to increased habitat homogenization and a subsequent reduction in beta diversity, several studies have rather observed similar or increasing beta diversity when comparing disturbed with undisturbed habitats (Hawkins et al. 2015; Petsch et al. 2021; Wang and Han 2023). Moreover, Pound et al. (2019) reported that heterogeneous stress contributed to beta diversity. Our findings fit this interpretation well, as our results could suggest that communities in two of the regulated lakes (Storvatnet and Stor-Drakstsjøen) experience higher stress than the third (Gjøljavatnet). However, we can expect that the eventual consequence of regulated lakes experiencing more severe stress or undergoing shifts in regulation regimes (e.g., due to increasing need for production flexibility) is decreased beta diversity. This pattern has been reported in different systems both by Pound et al. (2019) and Cortés-Guzmán et al. (2026), particularly due to loss of sensitive species in higher stress areas. This underlines the importance of considering how hydropower reservoirs may impact not only local, but also regional biodiversity.

Results from trait analyses were consistent with species replacement, and we observed a change in the prevalence of species with specific characteristics. Across control lakes, a diverse assortment of armor and locomotion trait types were represented (SI, Fig. S7). The significantly more common ones in controls were vegetation and silk case builders and temporarily attached individuals. We found nine unique

vegetation case building species in total (most abundant: *Annitella obscurata*, *Halesus radiatus*, *Lepidostoma hirtum*) and seven unique temporarily attached silk case builders (most abundant: *Polycentropus flavomaculatus*, *Cyrnus trimaculatus*, *Cyrnus flavidus*) across control lakes. Among the latter, Trichoptera species belonging to the family Polycnropodidae, which are temporarily attached predators constructing silk cases and capture nets, were abundant in all control lakes.

EPT communities in regulated lakes were overall less diverse (SI, Fig. S7), with a shift in trait composition. While the mineral armor trait type was not significantly more common in regulated lakes than in controls, the overall abundance of species with mineral cases was higher than in control lakes. These were represented by eight unique species across regulated lakes (most abundant: *Apatania stilgmattella*, *Limnephilus vittatus*, *Athripsodes cinereus*). Notably, in addition to the significant reduction in vegetation case builders across regulated lakes, vegetation-reliant case building species (both mixed-material and full vegetation armor) were completely absent from two of the regulated lakes. Temporarily attached silk case building species were also absent from both lake Stor-Drakstsjøen and Storvatnet with only low abundances in Gjøljavatnet of the three Polycnropodidae species most abundant in controls.

Other studies have indicated that several functional traits may be of relevance in explaining how affected a species is by regulation (e.g., uni- versus semivoltine). While we only selected two traits and excluded others, due to uncertain and lacking species-level trait values from literature and absence of required site-specific measurements, they may still play a role in explaining community changes following regulation.

While Aroviita and Hämäläinen (2008) found no taxa indicative of regulated lakes, some of the species indicative of reference lakes in their study were also found among the temporarily attached silk case builders in our study: *Cyrnus trimaculatus*, *Cyrnus flavidus* and *Polycentropus flavomaculatus*. A study from Wales (Hunt and Jones 1972) observed the loss of a *Leptophlebia* species along with most Trichoptera species following regulation. We similarly observed lower abundances of *L. vespertina* in regulated lakes. A decrease in low mobility groups was also observed by White et al. (2011). Heino (2008) reported that several invertebrate groups depend directly or indirectly on habitat structural complexity, and abundances of the functional groups these belonged to were significantly related to macrophyte cover. Similarly, a study of European lakes found structurally complex macrophyte habitats to have the highest macroinvertebrate taxon richness (Porst et al. 2019). In their study, Porst et al. (2019) were able to relate the changes in littoral communities to anthropogenic shoreline alterations, linking the effect to the loss of habitat and its complexity.

Another possible contributor to the reduced prevalence of vegetation case builders in regulated lakes could be reduced influence of riparian vegetation. The riparian zone provides structural complexity and organic material to littoral habitats and has been shown to influence benthic macroinvertebrate community composition (McGoff and Sandin 2012). Hydropower regulation can disrupt the connectivity between terrestrial vegetation and the aquatic habitat. This happens through the creation of a drawdown zone that is only periodically inundated, and that faces desiccation and freezing during winter drawdowns in boreal lakes (Carmignani and Roy 2017). This could lead to a reduced availability of shelter and plant material. Our field observations indicated that, in the regulated lakes, the waterline reached the terrestrial vegetation only at maximum water level (for visual comparison, see SI, Fig. S1). Reduced riparian input could therefore be contributing to the change in trait types. Examples of species found only in controls are *Limnephilus rhombicus*, *Mystacides azurea*, and *Glyptotendipes pellucidus*, which typically build protective cases with material of terrestrial/riparian origin such as leaves, twigs, and bark.

A mechanism explaining our findings could be that altered WLF has changed the state of the habitat, leading to a process of environmental filtering that indirectly impacts survival by determining which species can persist. The observed species turnover would accordingly be the result of a species replacement process where the altered environmental conditions would filter out certain species while allowing others to persist or establish. As known habitat alterations following regulation include the loss of vegetation, detritus, and fine sediments, it is possible that niche space has been reduced in the regulated lakes. This could be contributing to a reduction in abundance and species richness and explain why the species that dominate after regulation are morphologically and functionally more similar. Such a functional narrowing may reduce the community's capacity to maintain ecosystem functioning under additional stressors, making ecosystems less robust.

## Conclusions

Our results demonstrate significant differences in functional trait composition between regulated and control lakes, supporting an effect of hydropower regulation on littoral EPT communities. Patterns of reduced abundance, species richness, and differences in community composition were generally consistent with our hypotheses but received weaker statistical support and varied considerably among regulated lakes. While the limited sample size constrains statistical inference, consistent patterns across functional trait composition and descriptive community analyses, including  $\beta$ -diversity partitioning, suggest that hydropower regulation may contribute to the restructuring of littoral benthic communities, potentially through environmental filtering. We

presented varying hydropower impact intensity and lake characteristics (i.e., slope and macrophyte cover) as mutually nonexclusive explanations for the variability among regulated lakes. As study lakes were selected to be replicates of similar environmental conditions, a larger number of lakes along known gradients of hydropower impact are needed to disentangle these effects. There are two concrete implications of our findings for the monitoring and management of hydropower regulated reservoirs:

Impacts on EPT functional composition were detected even though the traditional metrics of abundance and richness were inconclusive. Reliance on richness-based assessments alone may therefore underestimate the regulation induced community degradation.

The loss of temporarily attached species appears to be a consistent signal of hydropower regulation impact even at small regulation amplitudes. Functional trait-based metrics may therefore offer a sensitive and mechanistically grounded indicator of regulation impact that could complement existing bioassessment frameworks such as the Water Framework Directive.

Our findings suggest that lake-specific characteristics should be considered when evaluating the ecological risks of hydropower regulation and prioritizing mitigation measures. Studies that cover broader gradients in both regulation regime and lake characteristics are needed to develop and validate new monitoring and management tools.

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**Data availability** The dataset can be accessed via the Global Biodiversity Information Facility (GBIF) (Hårsaker et al. 2025). The data and code supporting this study are openly available from Zenodo: 10.5281/zenodo.22829185 and 10.5281/zenodo.22830342, respectively.

## Declarations

**Conflict of interest** The authors have no relevant financial or nonfinancial interests to disclose.

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